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TECH GADGET STORE E-COMMERCE BACKEND API

1. Project Overview:

This project is a web-based E-Commerce Backend System used to manage customers, products, shopping carts, and orders. The system allows users to register, login securely, browse products, add products to carts, place orders, and perform various e-commerce operations through RESTful APIs. The backend is developed using Node.js, Express.js, MongoDB, and JWT Authentication following the MVC architecture.

2. Main Objectives:

- Secure user registration and login system
- JWT-based authentication and authorization
- Product management system
- Shopping cart management
- Order management system
- MongoDB database integration
- MVC architecture implementation
- API testing using Postman

3. User Roles:

Customer

- Register Account
- Login to System
- Browse Products
- Add Products to Cart
- Manage Cart Items
- Place Orders
- View Order History

Admin

- Manage Products
- Add Products
- Update Products
- Delete Products
- Monitor Orders
- Manage Inventory

4. Registration Flow:

Step 1: User registers using name, email, and password.

Step 2: Password is encrypted using bcrypt.

Step 3: User information is stored in MongoDB.

Step 4: User account is created successfully.

Step 5: User can login using registered credentials.

5. Login Flow:

User enters login credentials.

System validates email and password.

JWT token is generated.

User gains access to protected APIs.

6. Dashboard Structure:

Customer Dashboard

- View Products
- Add Products to Cart
- View Cart
- Update Cart Items
- Place Orders
- View Orders

Admin Dashboard

- Add Products
- View Products

- Update Products
- Delete Products
- Monitor Orders
- Manage Product Inventory

7. Form Workflow:

Product Workflow

Admin Login → Add Product → MongoDB Storage → Product Available for Customers

Cart Workflow

Customer Login → Browse Products → Add Product to Cart → Update/View Cart

Order Workflow

Customer Login → Create Order → Store Order in Database → View Order Status

Authentication Workflow

Register → Login → JWT Token Generated → Access Protected APIs

8. Functional Requirements:

- Secure Authentication
- User Registration
- User Login
- JWT Token Generation
- Product CRUD Operations
- Cart CRUD Operations
- Order CRUD Operations
- MongoDB Integration
- API Response Handling

9. Non-Functional Requirements:

- Security
- Password Hashing using bcrypt
- JWT Authentication
- Protected API Routes
- Performance
- Fast API Response
- Efficient Database Operations

- Scalability
- Expandable MVC Architecture
- Easy Feature Integration
- Support for Future Modules

10. Recommended Technology Stack:

- Backend
- Node.js
- Express.js
- Database
- MongoDB
- Mongoose
- Authentication
- JWT
- bcryptjs
- Testing
- Postman
- Git
- GitHub

11. Suggested Database Collections:

Users Collection

{name, email, password, role}

Products Collection

{name, brand, price, stock, category}

Cart Collection

{userId, productId, quantity}

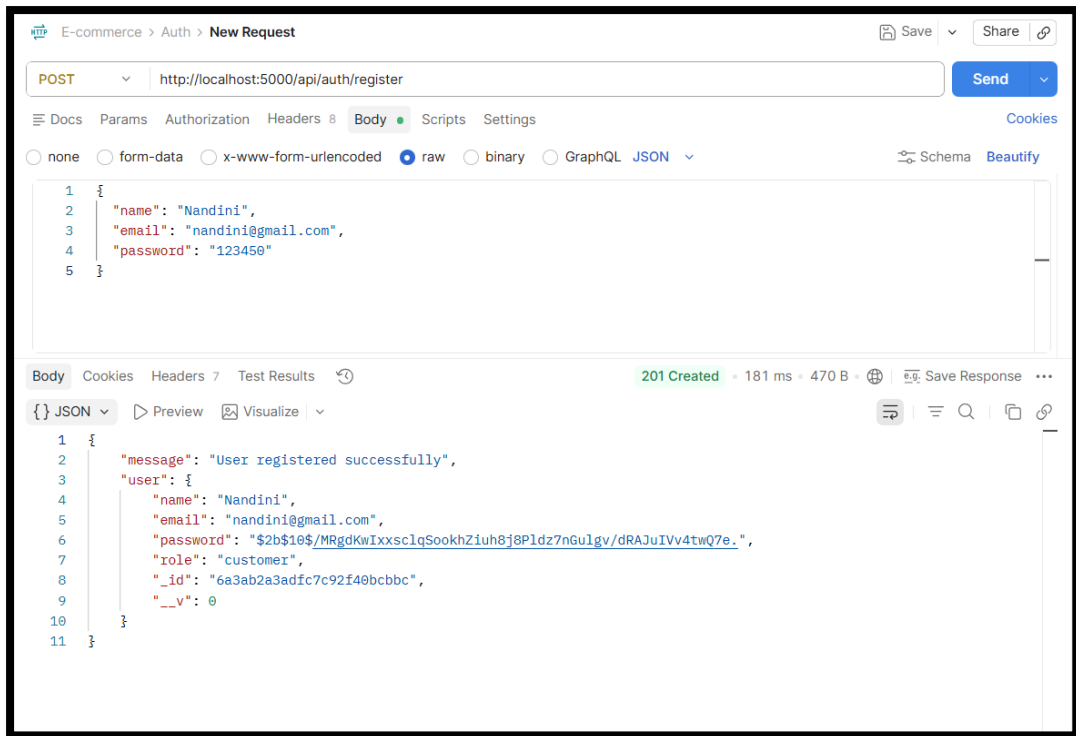
Orders Collection

{userId, products, totalAmount, status}

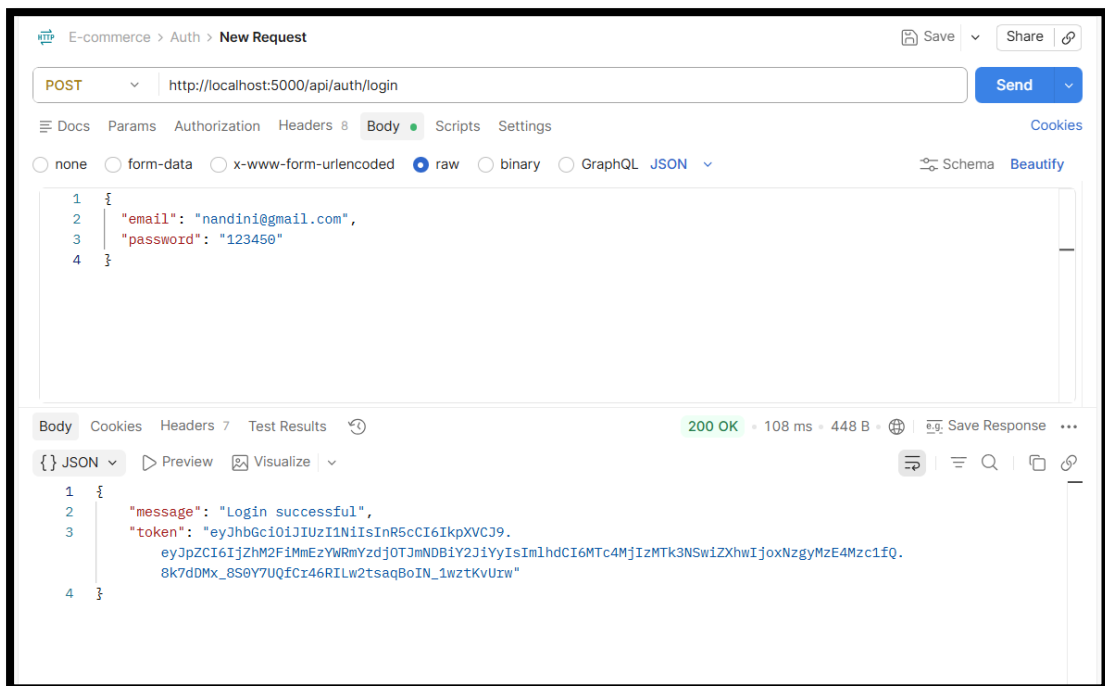
12. API Structure:

Authentication APIs

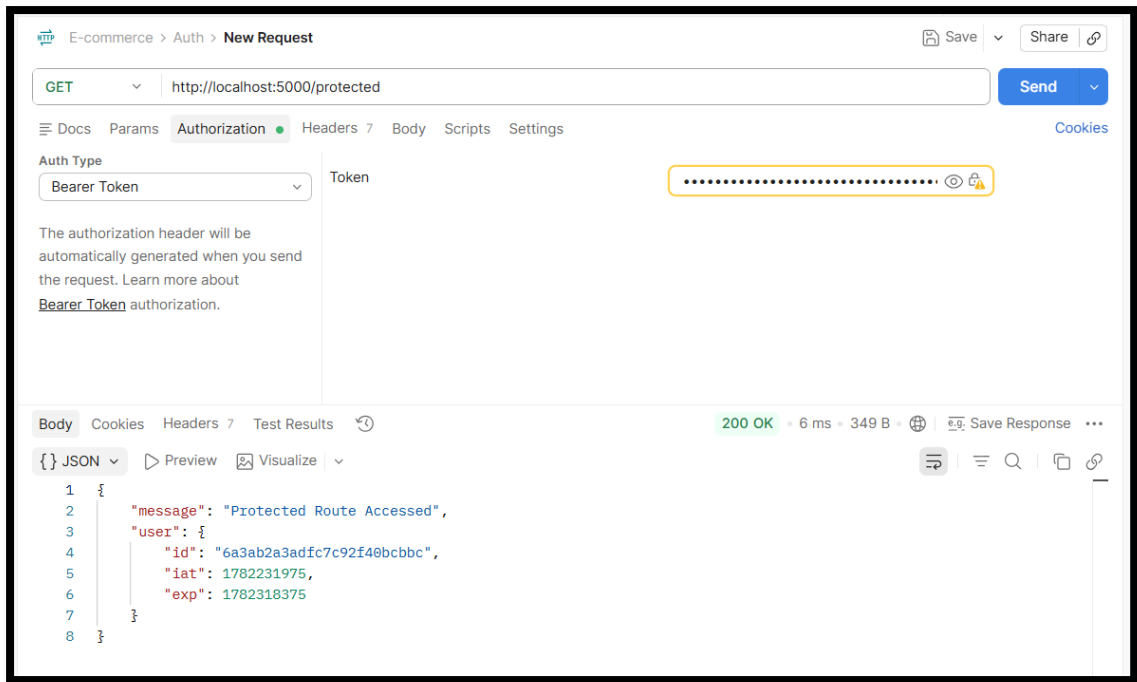
POST /api/auth/register



POST /api/auth/login

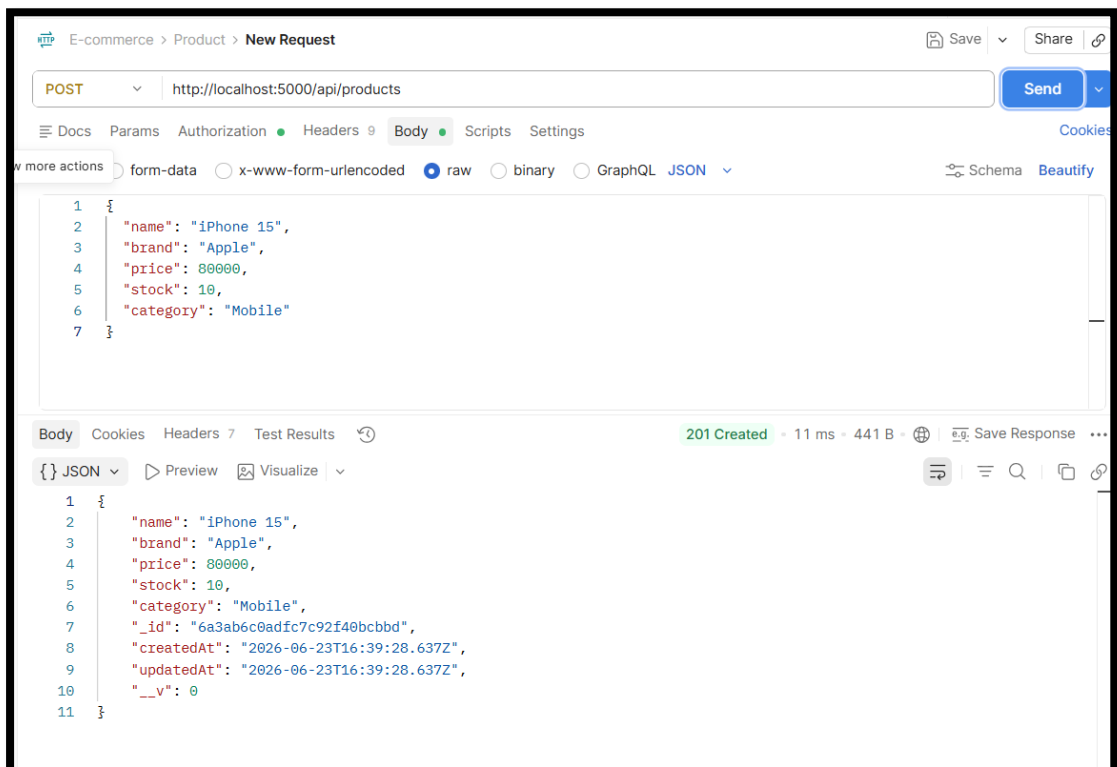


GET /protected

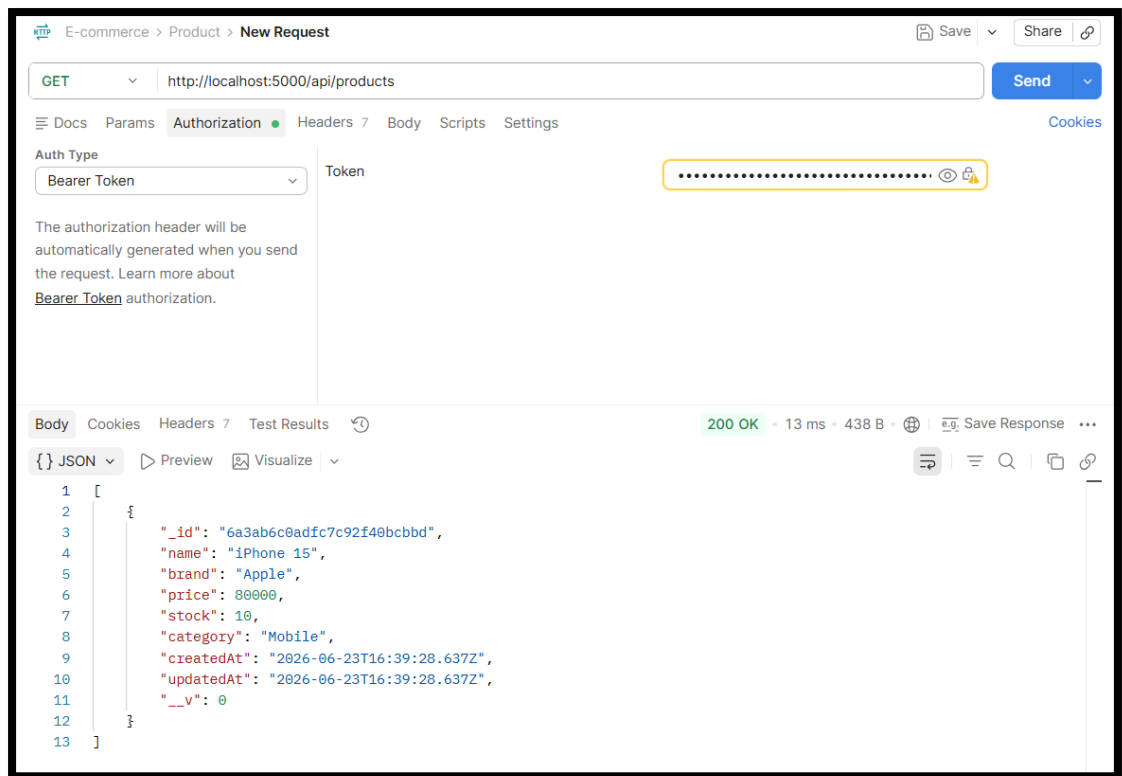


Product APIs

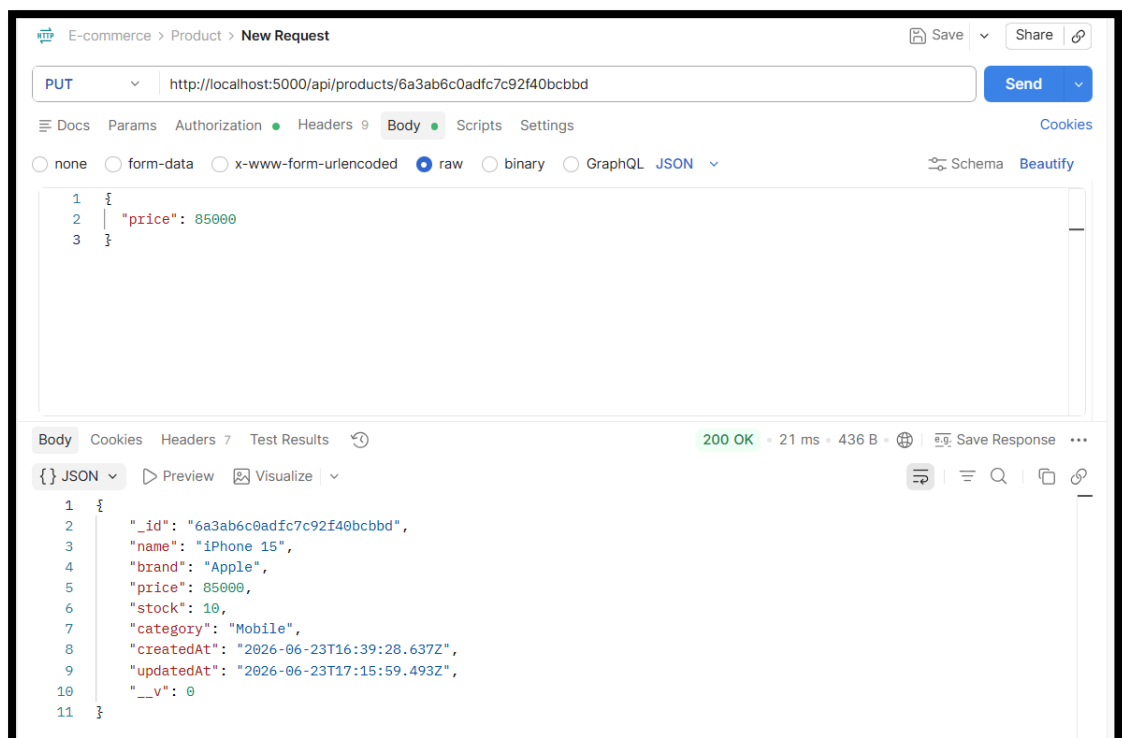
POST /api/products



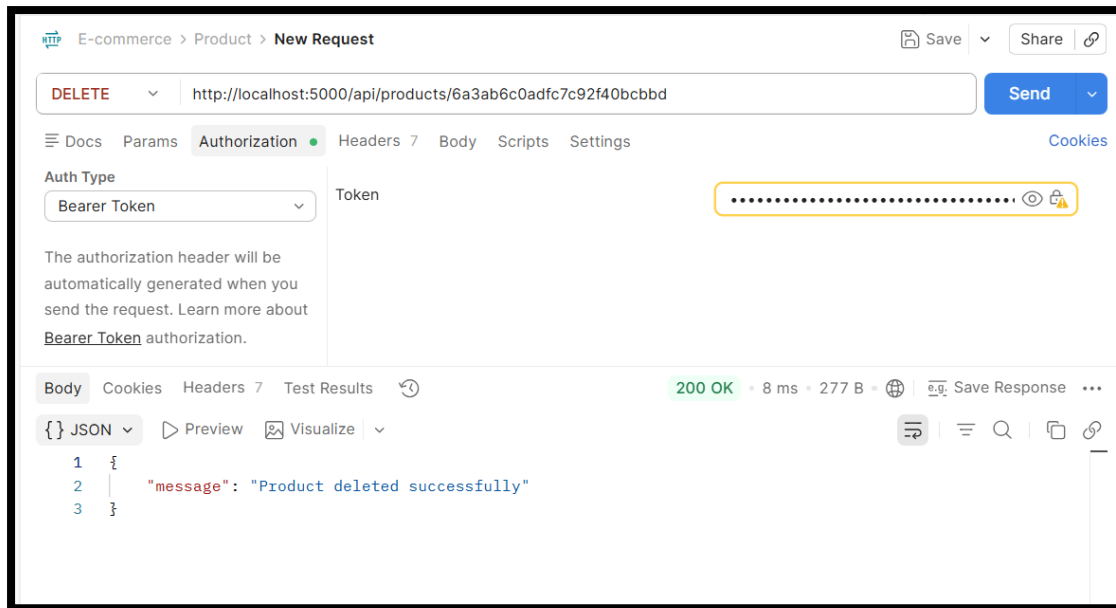
GET /api/products



PUT /api/products/:id

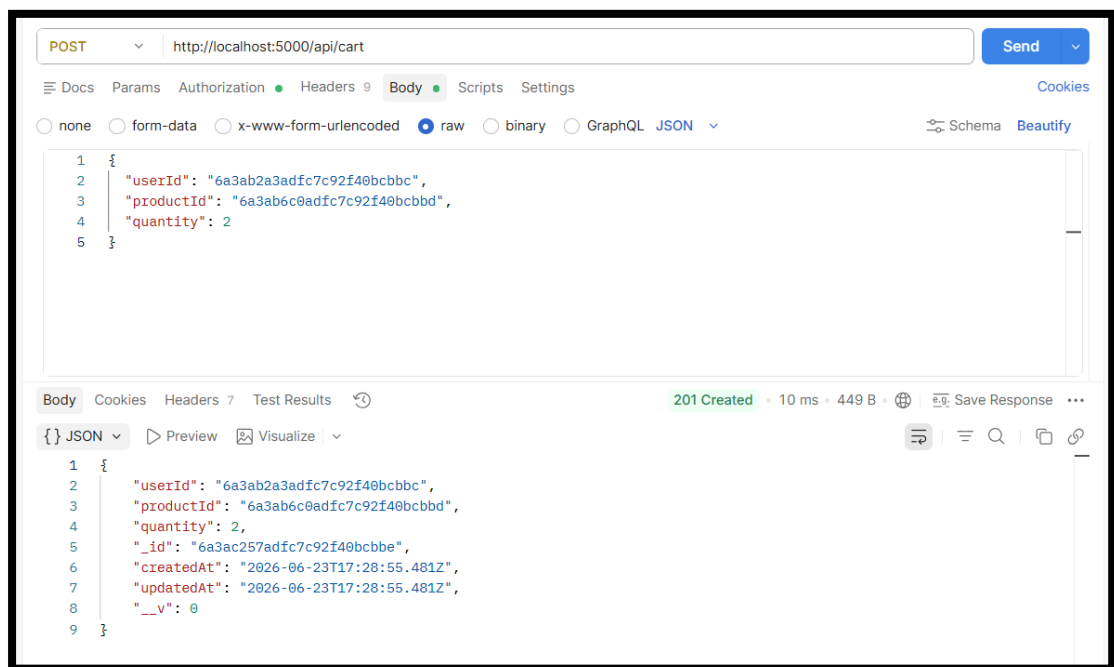


DELETE /api/products/:id

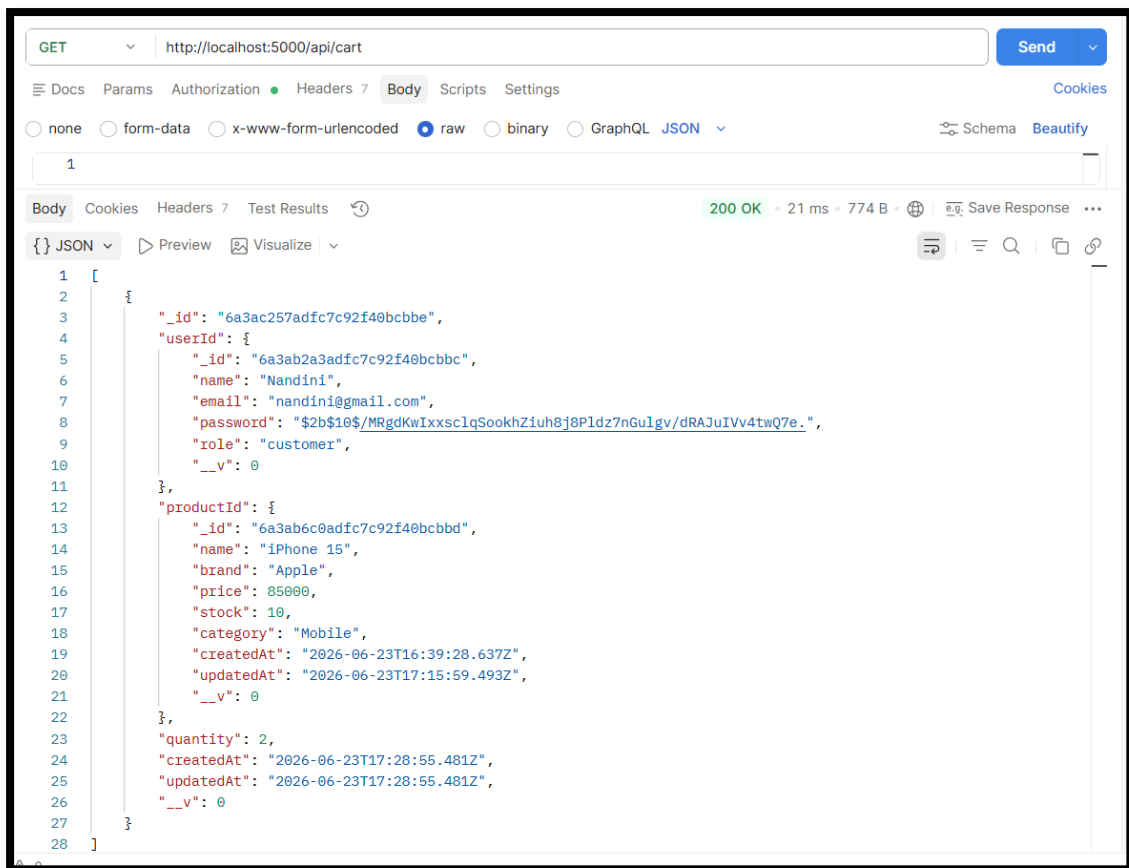


Cart APIs

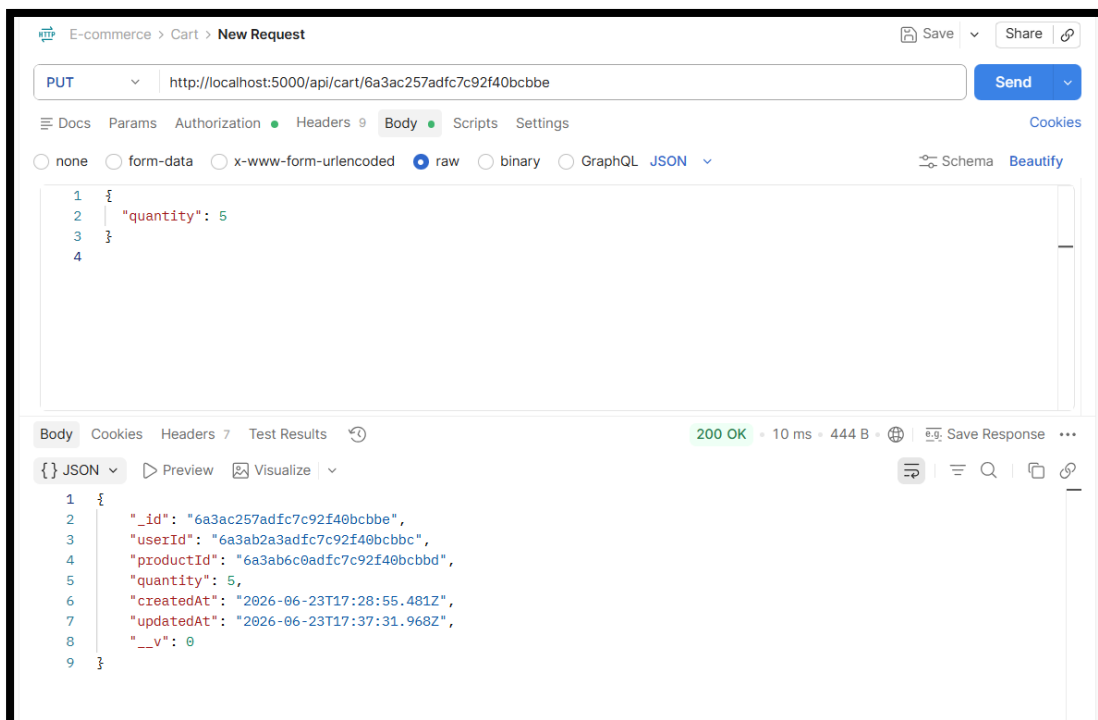
POST /api/cart



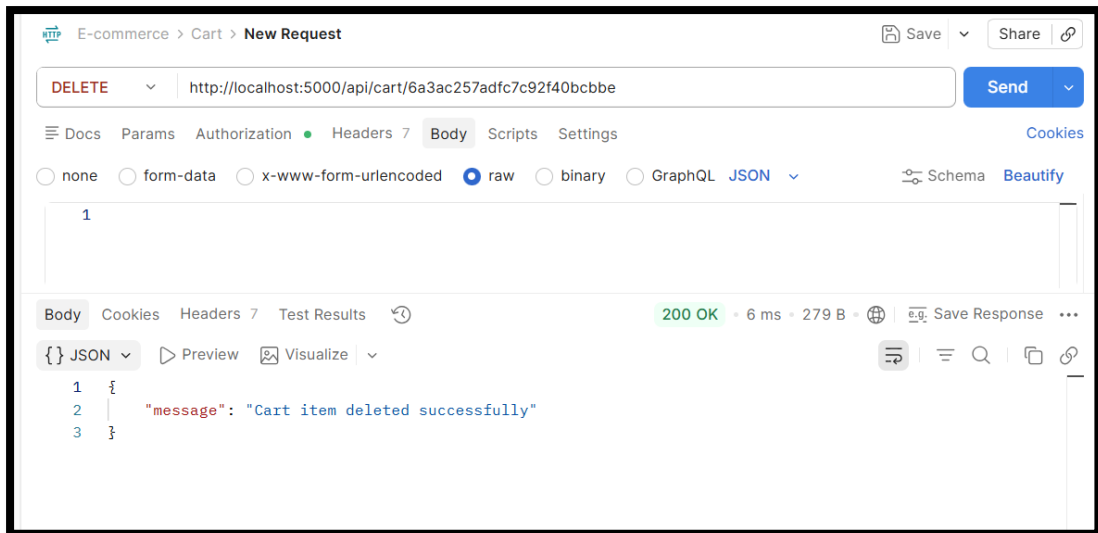
GET /api/cart



PUT /api/cart/:id

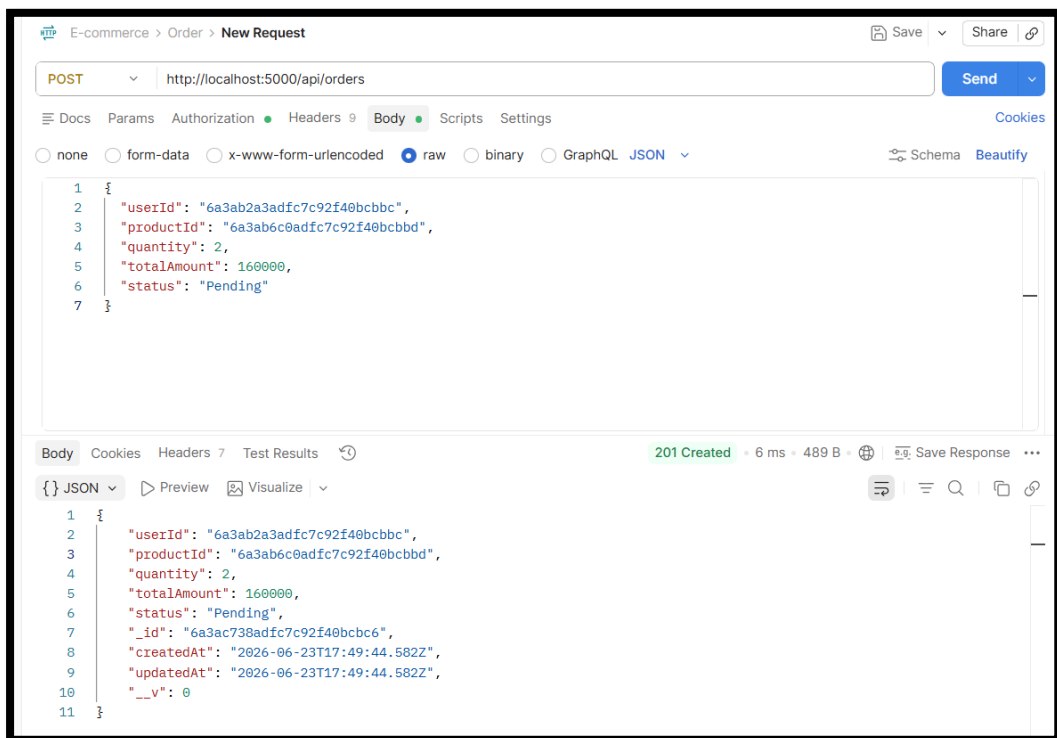


DELETE /api/cart/:id

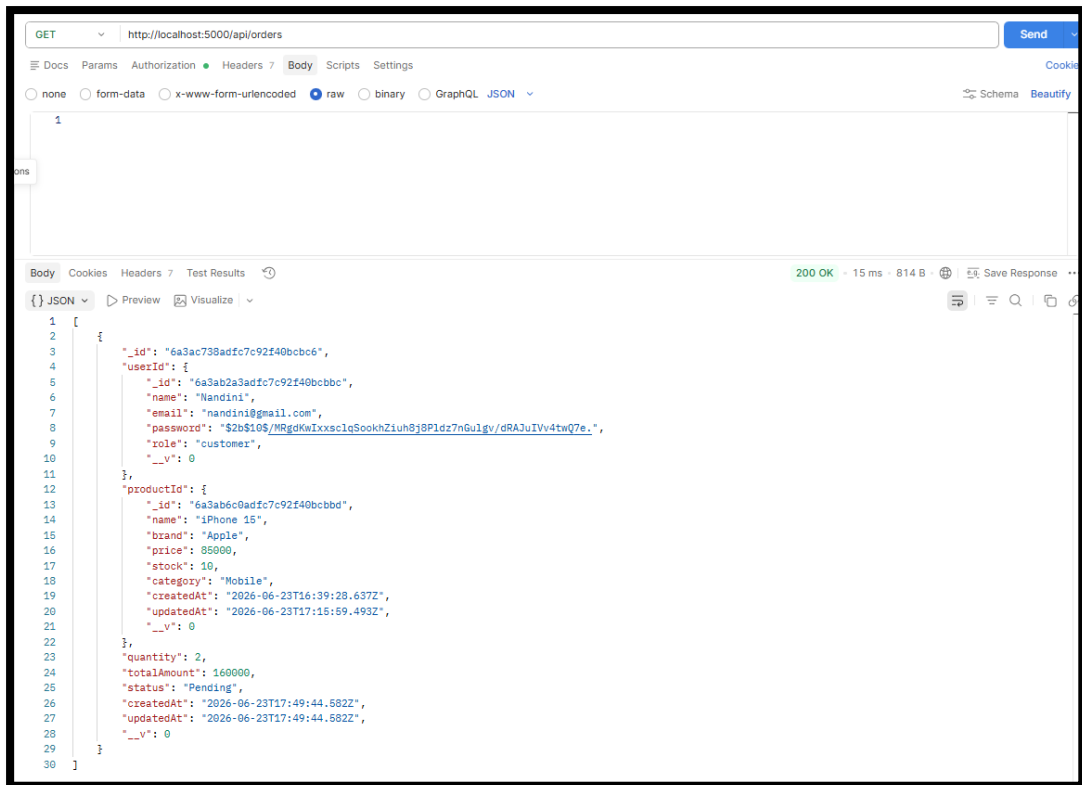


Order APIs

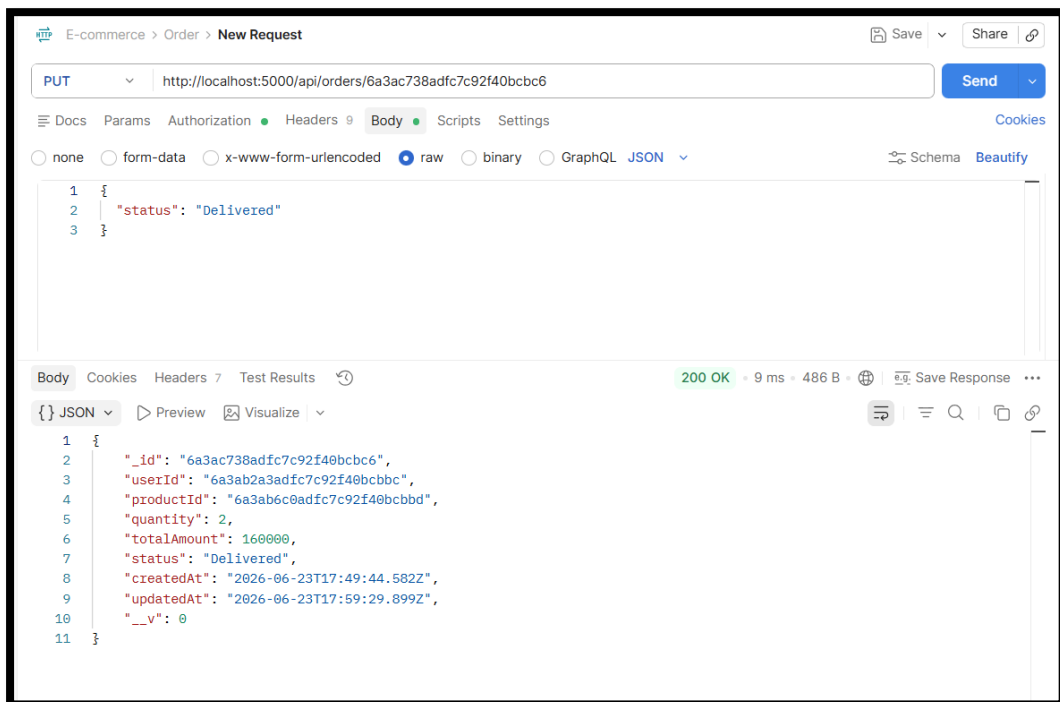
POST /api/orders



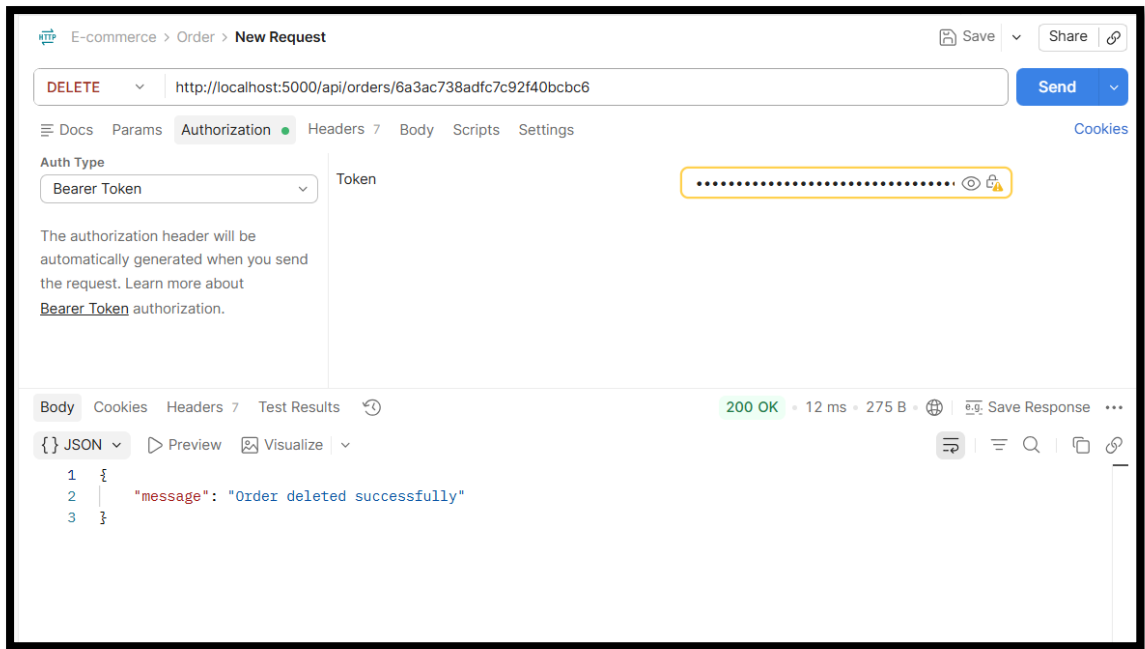
GET /api/orders



PUT /api/orders/:id



DELETE /api/orders/:id



13. Security Flow:

Login → JWT Token Generated → Authorization Header → Middleware Verification → Access Granted / Denied

14. Future Enhancements:

- Payment Gateway Integration
- Product Reviews and Ratings
- Wishlist Functionality
- Product Search and Filters
- Email Notifications
- Admin Dashboard
- Product Image Upload
- Order Tracking System
- Sales Analytics

15. Recommended Project Structure:

/server

/config

db.js

/controllers

authController.js

productController.js

cartController.js

orderController.js

/models

User.js

Product.js

Cart.js

Order.js

/routes

authRoutes.js

productRoutes.js

cartRoutes.js

orderRoutes.js

/middleware

authMiddleware.js

server.js

16. Development Phases:

Phase 1

- MongoDB Connection
- User Model
- Registration API

Phase 2

- Login API
- JWT Authentication
- Protected Routes

Phase 3

- Product CRUD Operations

Phase 4

- Cart CRUD Operations

Phase 5

- Order CRUD Operations
- Postman Testing
- GitHub Integration

Module:

Module	Collection
Authentication	users
Product Management	products
Cart Management	cart
Order Management	orders

Database Collections and Their Components

Tech Gadget Store E-Commerce Backend API

1. USERS Collection:

Stores registered user information.

Field Name	Data Type	Description
_id	ObjectId	Primary Key
name	String	User Name
email	String	User Email
password	String	Encrypted Password
role	String	customer/admin
createdAt	Date	Account Creation Date

2. PRODUCTS Collection:

Stores product information.

Field Name	Data Type	Description
_id	ObjectId	Primary Key
name	String	Product Name

Field Name	Data Type	Description
brand	String	Brand Name
price	Number	Product Price
stock	Number	Available Quantity
category	String	Product Category
createdAt	Date	Creation Date

3. CART Collection:

Stores shopping cart information.

Field Name	Data Type	Description
_id	ObjectId	Primary Key
userId	ObjectId	User Reference
productId	ObjectId	Product Reference
quantity	Number	Product Quantity
createdAt	Date	Creation Date

4. ORDERS Collection:

Stores customer order information.

Field Name	Data Type	Description
_id	ObjectId	Primary Key
userId	ObjectId	User Reference
products	Array	Ordered Products
totalAmount	Number	Total Order Amount
status	String	Order Status
createdAt	Date	Order Date

Main Collection Relationships

USERS

└─ CART

└── ORDERS

PRODUCTS

└── CART

└── ORDERS

17. Conclusion:

This project provides a complete E-Commerce Backend solution for managing customers, products, shopping carts, and orders. It implements secure authentication, JWT authorization, MongoDB integration, and RESTful APIs using the MVC architecture. The system is scalable, maintainable, and serves as a strong foundation for developing a full-featured e-commerce platform.